EXPERIMENT – INVESTIGATING INSULATORS

YOU WILL NEED (in pairs):

- 2 empty soft drink cans
- 2 thermometers
- newspaper
- polystyrene foam and sticky tape or foam drink can holder
- woollen cloth
- cottonwool
- foam rubber
- hot water
- measuring cylinder
- sticky tape (to tape on the materials)

STEPS

1. One of your cans will have no surrounding.
2. Surround your other can with the material you pair has been allocated.
3. Measure out and pour 100ml of hot water into each of the cans.
4. Measure the temperature of the water in each of the cans. Repeat the measurement of temperature every five minutes for 20 minutes.
5. Collect the temperature measurement for other surrounding materials of other groups in the class.
6. Record all you temperatures and the temperatures of other groups in the table below.

CAN COVERING	TIME (minutes)				
	0	5	10	15	20
None					
Cotton wool					
Woollen cloth					
Newspaper					
Polystyrene					

Draw a bar graph that will allow you to compare the drop in temperature of the water in the cans after 20 minutes.

QUESTIONS

1. Which covering appears to be the most effective insulator?

_____

2. Which one or more of the three methods of heat transfer does the most effective insulator reduce?

_____

3. Use your data to suggest a good container for a mug of hot coffee.

_____

4. Why was one can left without a covering?

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5. Are your conclusions reliable? Discuss the difficulties encountered in making sure that the comparison of insulators was fair.

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